

Computer Graphics Semester Project

Project Principle

Don't just build something that looks graphical. Build something that demonstrates that you understand how computer graphics work.

1. Project Overview

Throughout the semester, you will work in groups of up to 6 students to design and develop a computer graphics application.

The project will be developed progressively throughout the semester. You are not expected to complete the entire project at once. Instead, each stage of the course will introduce graphics concepts and techniques that you will apply to your project.

The purpose of the project is to give you experience with:

- Applying mathematical and geometric concepts in computer graphics
- Understanding and implementing graphics algorithms
- Working with a graphics API
- Developing a graphical application from an initial idea to a working implementation
- Working collaboratively as a development team
- Planning and managing a project over an extended period
- Explaining the graphics techniques used in your application

Important!

Your project does not have to be a game.

You are encouraged to explore different applications of computer graphics, provided that your project requires meaningful use of computer graphics concepts.

Examples could include:

- Games or interactive entertainment
- Simulations
- Scientific or mathematical visualisations
- Educational applications
- Drawing or design tools

- Data visualisations
- Architectural or 3D visualisations
- Interactive demonstrations
- Virtual environments
- Creative/artistic applications
- Other applications that make meaningful use of computer graphics

The focus is **not on how complex or flashy the application looks**, but on how effectively you apply and explain computer graphics concepts.

2. How the Project Will Develop

Your project will follow four broad phases.

Phases	Expectations
Phase 1 - Concept & Design	You will define what you want to create and identify the computer graphics concepts that may be required.
Phase 2 - Core Graphics	You will implement fundamental graphics concepts such as geometric primitives, drawing and algorithms
Phase 3 - Advanced Graphics	You will progressively incorporate more advanced techniques introduced during the course such as transformations.
Phase 4 - Integration & Refinement	You will combine the techniques into a coherent application, test and refine your implementation, and prepare the final project.

Your project is expected to evolve.

Your initial proposal does not need to contain every technical detail of the final implementation. As you learn new techniques, you may refine your project and its implementation.

3. Project Requirements

Your final project must demonstrate meaningful use of computer graphics techniques covered in the course.

Depending on your project, this may include; geometric primitives, line and circle drawing algorithms, polygon operations, curves, 2D transformations, 3D transformations, raster operations, animation, color techniques, lighting and shading, texture mapping and other appropriate graphics techniques

You are not required to use every technique taught in the course.

Instead, your group should be able to justify **why this graphics technique is appropriate for our project** and **how does our implementation use the technique**

4. Group Responsibilities

Although the project is completed as a group, **every member must contribute to the project.**

At the beginning of the project, assign members initial responsibilities.

Possible responsibilities include; graphics/algorithm implementation, interface/application development, 2D/3D modelling, animation, graphics effects, testing, documentation and integration

These responsibilities are **not fixed**. Members should collaborate and understand the overall project rather than working entirely independently.

Each member should be able to explain their contribution and how it fits into the overall application.

4.1 Member Contribution

Each group member's contribution to the project will be tracked through the group's **GitHub repository**.

Members are expected to contribute regularly by committing and pushing their work to the repository throughout the semester. GitHub activity will be used as evidence of individual participation and contribution to the group project. Therefore, **members should ensure that their work is committed under their own GitHub account and that contributions are made consistently rather than submitted in bulk at the end of the project.**

5. Development Approach

The project will be developed incrementally.

You should avoid leaving the majority of the project until the end of the semester.

At different points in the semester, your group will be expected to demonstrate progress such as planning, implementation, testing, integration and refinement

Each new graphics technique should, where appropriate, provide an opportunity to improve or extend your project. This means that the **weekly practical activities are not separate from your project**. They are intended to help you build the skills required to complete it.

6. Technology

The project will use the graphics programming environment specified for the course.

Your group will maintain a **GitHub repository** containing your project source code and relevant documentation.

Your repository should be organised so that another person can understand:

- What the project does
- How to run it
- What technologies were used
- Who contributed to the project
- What graphics techniques were implemented

Further technical requirements will be provided as development progresses.

7. What Makes a Good Computer Graphics Project?

A strong project should demonstrate:

- **A clear purpose;** The application should have a reason for existing beyond simply displaying shapes.
- **Meaningful graphics;** Computer graphics should be central to the application, not just decoration.
- **Appropriate use of techniques;** Graphics algorithms and techniques should be selected because they serve a purpose within the project.
- **Progressive development;** The project should show evidence of development throughout the semester.
- **Understanding;** Your group should be able to explain the mathematical concepts, algorithms and implementation decisions behind your graphics.
- **Collaboration;** All group members should make meaningful contributions.
- **Scope control;** A smaller, well-implemented project is preferable to an ambitious project that is incomplete.